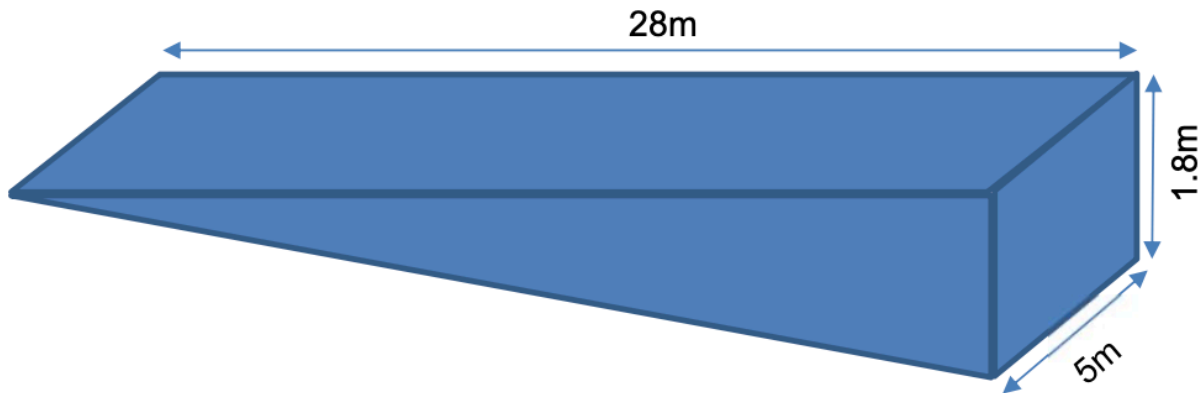


⚠ REPRODUCTION OF THIS MATERIAL IS STRICTLY PROHIBITED

A tourist resort's swimming pool is made from a triangular prism as shown in the picture below. The sloping bottom of the pool makes a 4° angle with the horizontal.



Given the pool's dimensions above and that a hose is filling up the pool at a constant rate of $0.125m^3min^{-1}$, at what rate is the area of the surface of the water increasing when the height of the water reaches $1.2m$?

In a secret Uranium mine in the outback of Australia, the Uranium ore is coming off the end of a conveyor belt and falling into a pile that can be modelled by an inverted cone. The rate at which Uranium ore is moving along the conveyor is $2.6m^3$ per minute.

The properties of Uranium ore are s.t. the resulting cone has the same height as the diameter. Use this information to find the rate of increase of the sloped height when the base area of the cone is $16\pi m^2$.

Prove the quotient rule using first principles:

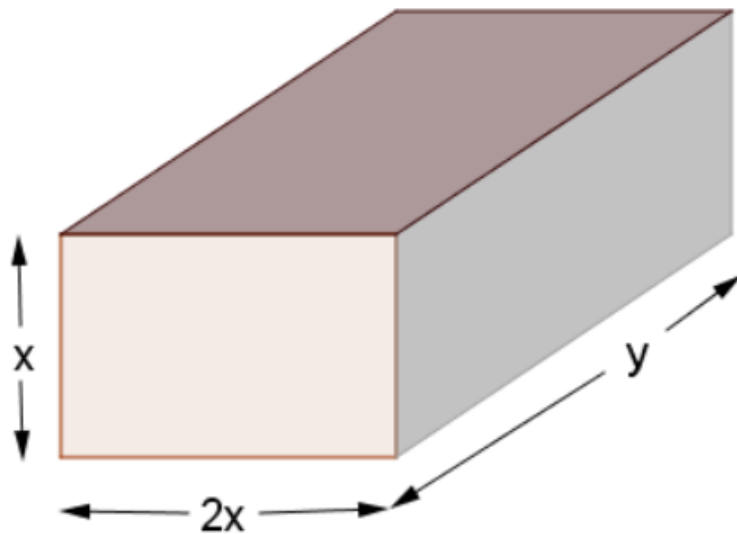
If

$$f(x) = \frac{g(x)}{h(x)}$$

Then

$$f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{[h(x)]^2}$$

A rectangular box (cuboid) with sides of different lengths, as shown in the diagram, and has a volume of



$32m^3$.
the cuboid s.t. the surface area is a minimum?

What are the dimensions of

A function is defined parametrically as

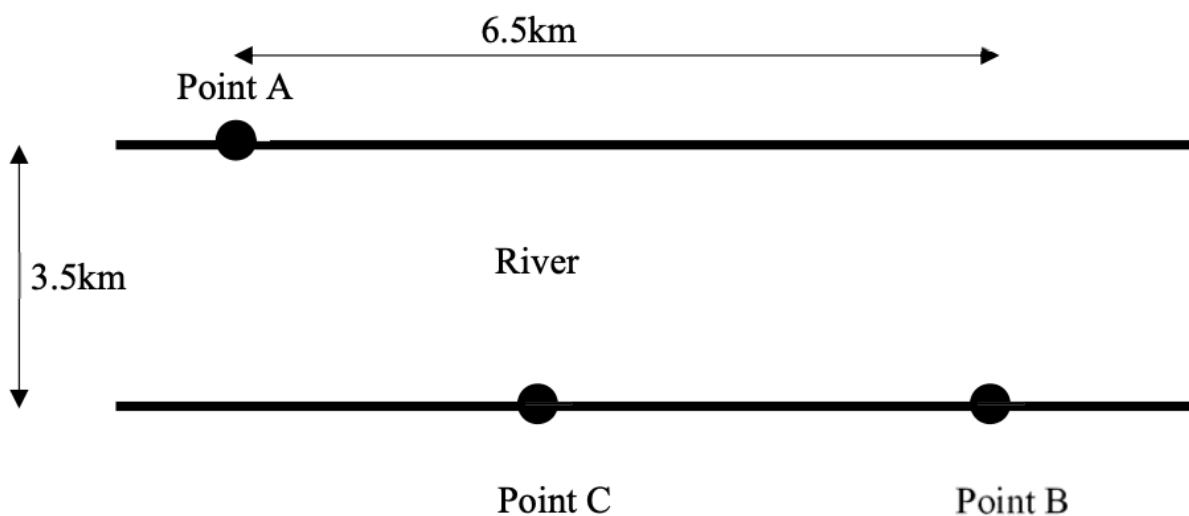
$$x = e^{\frac{2}{\pi}t-2} \sin t$$

and

$$y = 6 \cos \left(\frac{1}{2}t \right)$$

Find the equation of the tangent of the function at $t = \pi$.

Francesca is in a multisport race and is planning her river crossing. She must paddle her kayak from point A on one bank of a $3.5km$ wide straight river and can choose where she exits on the far bank before making her way to point B, which is $6.5km$ downstream.



If she can paddle at $5kmh^{-1}$ and run at $8kmh^{-1}$, find the point C on the far bank that will allow her to get to point B as quickly as possible.

Sand is being poured into a very large container at a rate of $0.3m^3$ per second. The interior of the container is in the shape of a cuboid with a base area of $4m^2$. This particular sand exerts a force (in Newtons), on the center of the base of the container equal to $F = 25h^2$, where h is the height of the sand in meters. Find the rate at which the force is increasing when there is $14m^3$ of sand in the container.

An object's velocity can be modelled, at any time (minutes) after starting, by

$$v(t) = \frac{t-1}{e^t}$$

Find the maximum velocity reached by this object within its first 6 minutes of travel.

A curve is defined by

$$y = (x-1)(x+3)$$

Given the tangents to this curve through the points $P(1,0)$ and $Q(x_1, y_1)$ meet at a right angle, determine the location of Q .

For a parametrically defined curve

$$x = \ln(t)$$

and

$$y = \cos(2t)$$

Find the gradient and concavity at the point when $t = \pi$.

If the function

$$f(x) = x^3 - 3x^2 + kx - 5$$

has a root at $x = -1$, show that there is also a stationary point. Determine this stationary point's nature.

A curve is defined parametrically by

$$x = 2 \tan(t)$$

and

$$y = 4 \sec(t)$$

Find the concavity of the curve when $t = \frac{\pi}{4}$

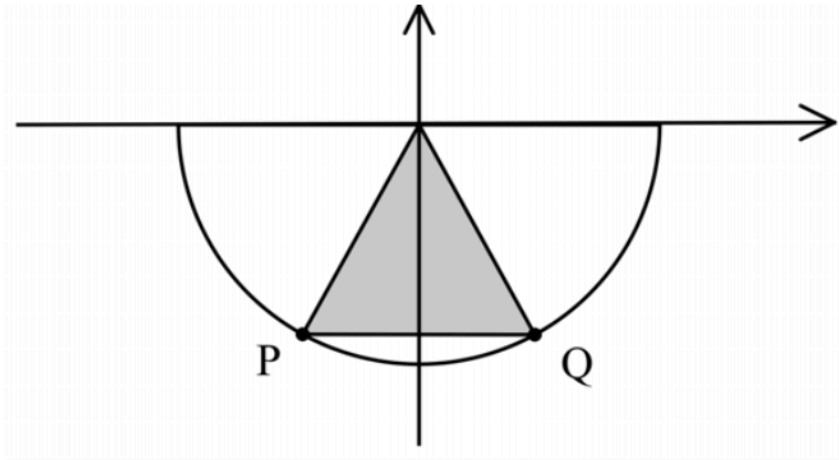
If

$$y = e^{-x} \sin(kx)$$

where k is a constant, find the value of C , if

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + ke^{-x} \cos(kx) = Cy$$

A triangle is inscribed into a semi-circle with radius r , where one vertex lies at the centre of the circle and the other two vertices, P and Q , lie on the semicircle as shown in the diagram below.



Find the height and base of the triangle when the area is a maximum.

Consider the curve

$$y = \frac{\pi \cdot \sin x}{x}$$

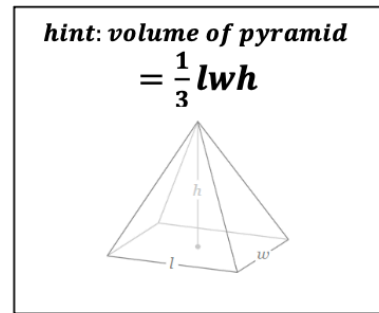
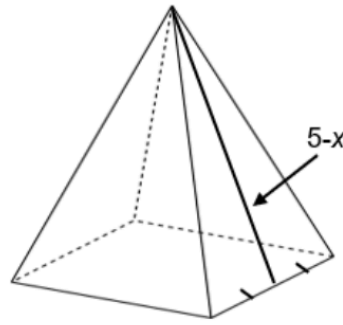
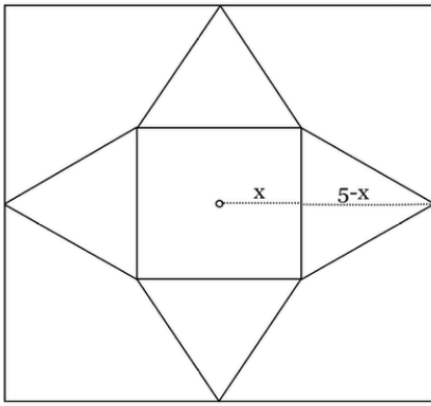
where $x \neq 0$, and its tangents at $x = \pi$ and $x = -\pi$.

Find the coordinates of the point where these two tangents intersect AND show that they meet at right angles there.

A tall street light is $6.5m$ above the flat ground. Matiu, who is $1.5m$ tall, is walking towards the point directly below the streetlight at 2 metres per second.

At what rate is the length of his shadow, L , changing when his distance, x , is $8m$ away from the point directly under the light?

Jan has a $10m$ by $10m$ square of canvas tent material which she will use to cut out a shape that will form a tent for her grandchildren.



The shape she cuts out is a square surrounded by 4 identical isosceles triangles as shown in the diagram. The tent will be a pyramid with a square base, as shown below. Use calculus to find the maximum volume.

The parametric equations of a curve are

$$x = e^t(1 + t^2)$$

and

$$y = e^t(1 - t^2)$$

where $t \in \mathbb{R}$, $t \neq -1$.

Find the equation of the normal to the curve when $t = 1$.

Find the turning points of the curve

$$f(x) = \frac{(2x - 3)^2}{4(x^2 - 1)}$$

where $x \neq \pm 1$.

Then, state the nature of the turning points.

Given that $e^y = x^3 \cos^2 x$, where $x > 0$, if

$$\frac{dy}{dx} = \frac{a}{x} + b \tan x$$

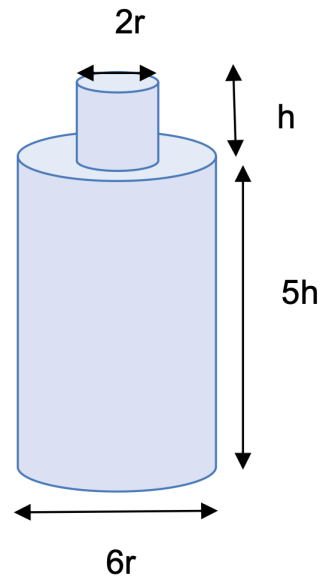
where a and b are constants, find the values of a and b .

A curve has the equation $y = 2e^{-x}$.

Prove that

$$\frac{d^2y}{dx^2} - \frac{1}{2} \cdot \frac{dy}{dx} - \frac{3}{2}y = 0$$

An open vinegar bottle which has the volume of $391\pi \text{ cm}^3$ is to be designed with two cylindrical shapes is given below. The first cylinder has the diameter of $2r \text{ cm}$ and the height of $h \text{ cm}$. The second cylinder has



the diameter of $6r \text{ cm}$ and the height of $5h \text{ cm}$.
Find the dimensions of the bottle which give the minimum surface area.

Find the dimensions of the

A metallic cube is being uniformly heated, and as such it is expanding uniformly. The surface area of the cube is found to be increasing at a constant rate of $0.5 \text{ cm}^2 \text{ s}^{-1}$.

Find the rate at which the volume of the cube is increasing when the cube's side length is 7 cm .

Find the coordinates of the minimum point of the function

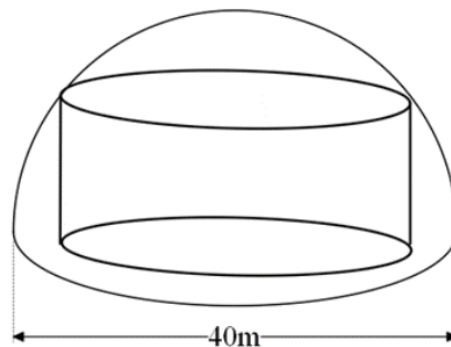
$$f(x) = \sqrt{2x} + 2\sqrt{k-x}$$

in terms of k , where $k > 0$, $k \in \mathbb{R}$.

A 3 m ladder is leaning right up against a wall. A gust of wind causes the bottom of the ladder to slip out from the wall at a rate equal to $\frac{10}{h}$ metres per second, where h is the height of the top of the ladder in metres. The top stays in contact with the wall as it is heading towards the ground.

Find the rate at which angle the ladder makes with the ground is changing when the base of the ladder is 1.2 m from the wall.

A hemi-spherical dome is to have a cylindrical tank built inside it. The cylindrical tank will be built so that its top edge touches the hemi-spherical dome. If the dome is 40 m in diameter, then what are the



dimensions that will maximize the volume of the cylinder?

Consider the parametrically defined curve

$$x = e^t$$

and

$$y = \frac{e^t}{t}$$

By finding the second derivative, describe the nature of the curve at $t = 2$.

A spherical balloon has been inflated but is leaking air at a constant rate of 8cm^3 per second. Find the rate at which the surface area of the balloon is decreasing when there is 10500cm^3 of air left in the balloon.

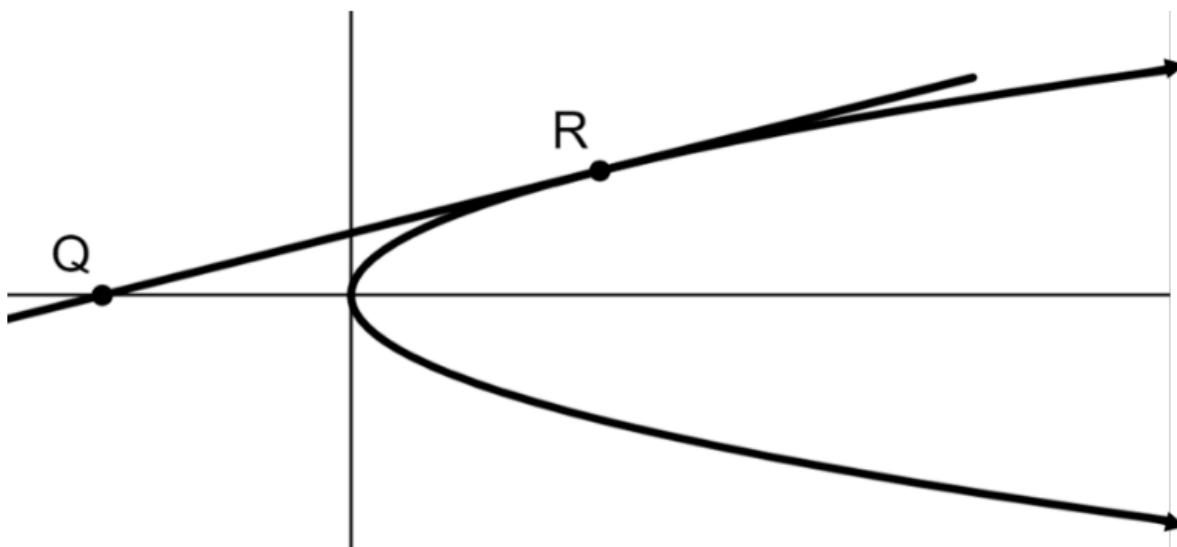
The equation of a parabola can be given parametrically by the equations

$$x = at^2$$

and

$$y = 2at$$

Any tangent to the parabola crosses the x -axis, as shown below. Show that the y -axis bisects the line segment QR , where Q is the x -intercept and R is the intersection of the parabola and its tangent.

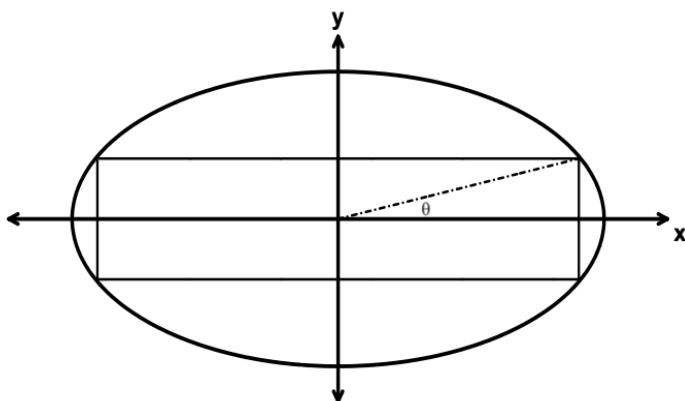


As shown in the diagram below, a rectangle is inscribed in an ellipse with parametric equations

$$x = 20 \cos \theta$$

and

$$y = 10 \sin \theta$$



the area of the rectangle.

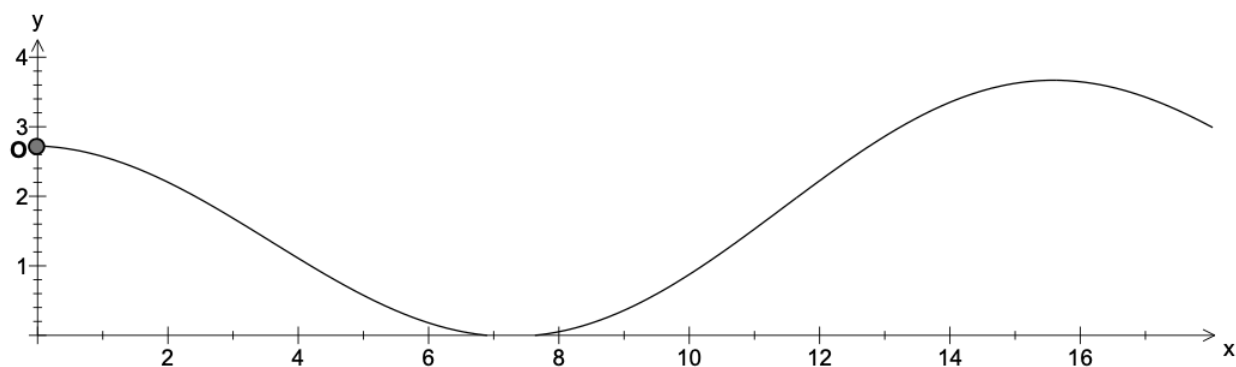
Find the value of θ that maximizes

The Crossans are looking at purchasing a section of land to build on. They do not want to buy land if the slope at the steepest part of the land is greater than 0.5.

The cross-section of this land can be modelled by the equation

$$y = -1.6 \cos(0.4x - 3) + 0.06x + 1.14$$

where x is the horizontal distance from the left point O and y is the height of the land above the lowest point on the section.



Determine, using calculus methods, where this land is steepest and whether you would recommend for the Crossans to buy the land.

For the function

$$f(x) = \frac{1}{x}$$

show that

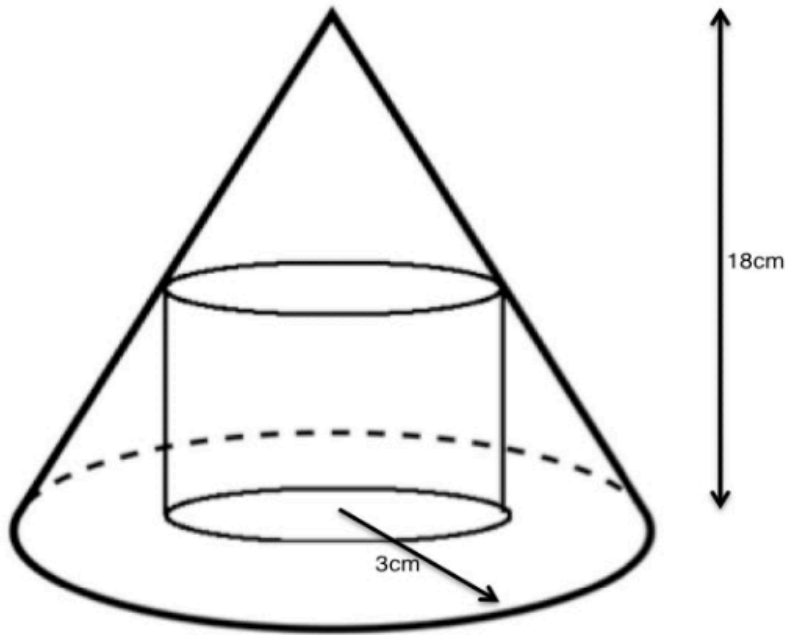
$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

gives $f'(x)$.

Find any inflection point(s) on the curve defined by

$$y = (x - 1)e^{x^2 - 1}$$

A cylinder is placed inside a cone as shown below.



What height must the cylinder have if its volume is to be a maximum?

Consider the function

$$f(x) = \tan(3x^2) - 3x^2$$

A line L is normal to the graph of $y = f'(x)$.

Show that the line L is parallel to the y -axis at $x = \sqrt{\frac{\pi}{3}}$.

Consider the curve

$$y = \ln(e^x + e^{-x})$$

Explain whether the curve has any point(s) of inflection AND account for the concavity of the curve.

Sam is doing an aviation training programme, flying from point A to point D . The distance between A and D is 20 kilometres. Points A , P , Q , and D lie in a straight line on the horizontal ground. The distances AP

and QD are each 1 kilometre. He is instructed to level off at a point PT and begin his ascent at a point on TQ , where $PT = TQ$ and PT is perpendicular to TQ .

After Sam takes off at point A , he climbs in a straight line directly to point B on PT . After reaching point B , the plane levels off and flies horizontally through an air corridor to point C on TQ . From point C , it descends in a straight line directly to point D .

The plane travels at 280 kilometres per hour on the diagonal paths AB and CD and 200 kilometres per hour

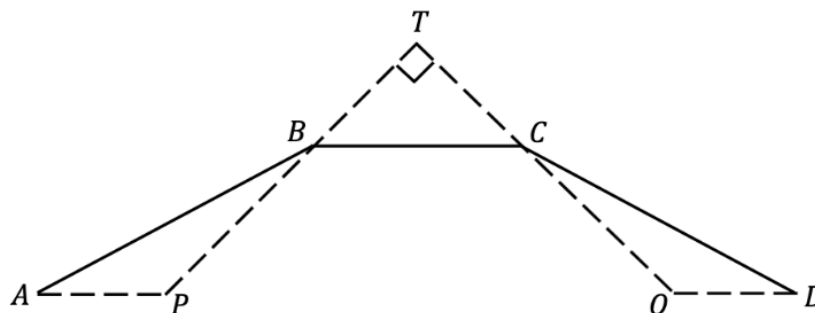


Diagram is NOT to scale

on the horizontal air corridor BC .

Find the minimum flight time from point A to point D .

Consider the function

$$f(x) = e^{-x}(x^2 + kx + 1)$$

where k is a real constant. The curve has exactly one stationary point.

A student claims that there might be a value of k s.t. the curve has no points of inflection.

Show that this claim is false. That is, show that the curve always has points of inflection.

In a game of table tennis, Aroha hits the ball towards Ben. The ball is travelling in a straight line trajectory. Aroha hits the ball from 1 metre above the ground, at an angle of elevation θ , where $0 < \theta < \frac{\pi}{2}$. To return the shot, Ben is forced to move backwards. Based on the speed and spin of the shot, the horizontal distance x , in metres, between the two players at the moment Ben attempts to intercept the ball can be modelled by the equation

$$x(\theta) = 3 \sin \theta \cos^2 \theta$$

Ben can reach a maximum vertical height of 2.2 metres with his paddle.

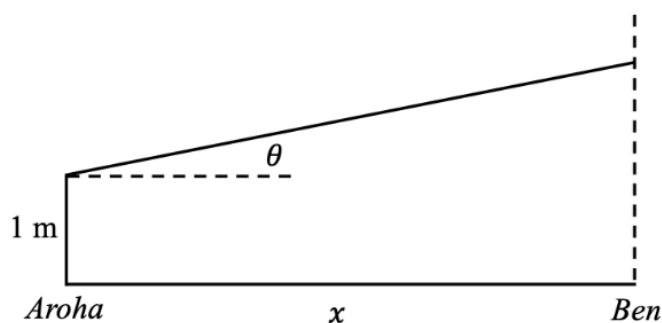


Diagram is NOT to scale

Determine whether it is possible for Aroha to hit the ball at an angle θ s.t. it passes completely over Ben's maximum reach.

Consider the functions

$$f(x) = 2x^2 - kx + 3$$

and

$$g(x) = \frac{k}{x}$$

where $x \neq 0$ and k is a real constant.

If $f(x) > g(x)$ for all values of x when $1 \leq x \leq 2$, show that $k < \frac{5}{2}$.

Consider a curve defined by the pair of parametric equations:

$$x = 27 \cos^3 t$$

and

$$y = 27 \sin^3 t$$

where $0 \leq t \leq \frac{\pi}{2}$.

Tangent line to this curve, intersects the x -axis at point A and the y -axis at point B . Shows that the distance between point A and point B is exactly 27.

A manufacturer is testing a new cylindrical juicer. The juicer consists of an outer cylinder of radius $3r$ cm and the solid central cone of radius r cm. The cylinder and the central cone have identical heights, which are four times the radius of the central cone. During a test, lemon juice is continuously poured into the juicer at a constant rate of $10\text{cm}^3\text{s}^{-1}$.

Show that the rate at which the depth of the juice is increasing at the exact moment the juice, reaches halfway up the juicer is $\frac{8}{7\pi r^2}\text{cm}^{-1}$.

An isosceles trapezium has a higher base length of two and its two equal non-parallel sides, each have length two as shown in the diagram. Find the interior angle, θ between the lower base and one of the equal

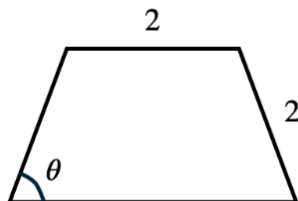


Diagram is NOT to scale

sides, which will maximize the area of the trapezium. You must justify that the area you have found is indeed a maximum.
